

Worksight

DESERT EAGLES · AMD DEVELOPER HACKATHON ACT III

Intelligent Industry track · visual quality inspection

Find defects on a new part from 20 photos of good parts. Gemma on AMD explains what to check.

web-production-d9743.up.railway.app



Why this project

A new part on the line has no defect dataset.

Defects are rare, so collecting and labelling hundreds of them takes weeks, and every new part or camera view starts again.

Quality engineers end up checking by eye. Worksight starts from what every line already has: photos of good parts.

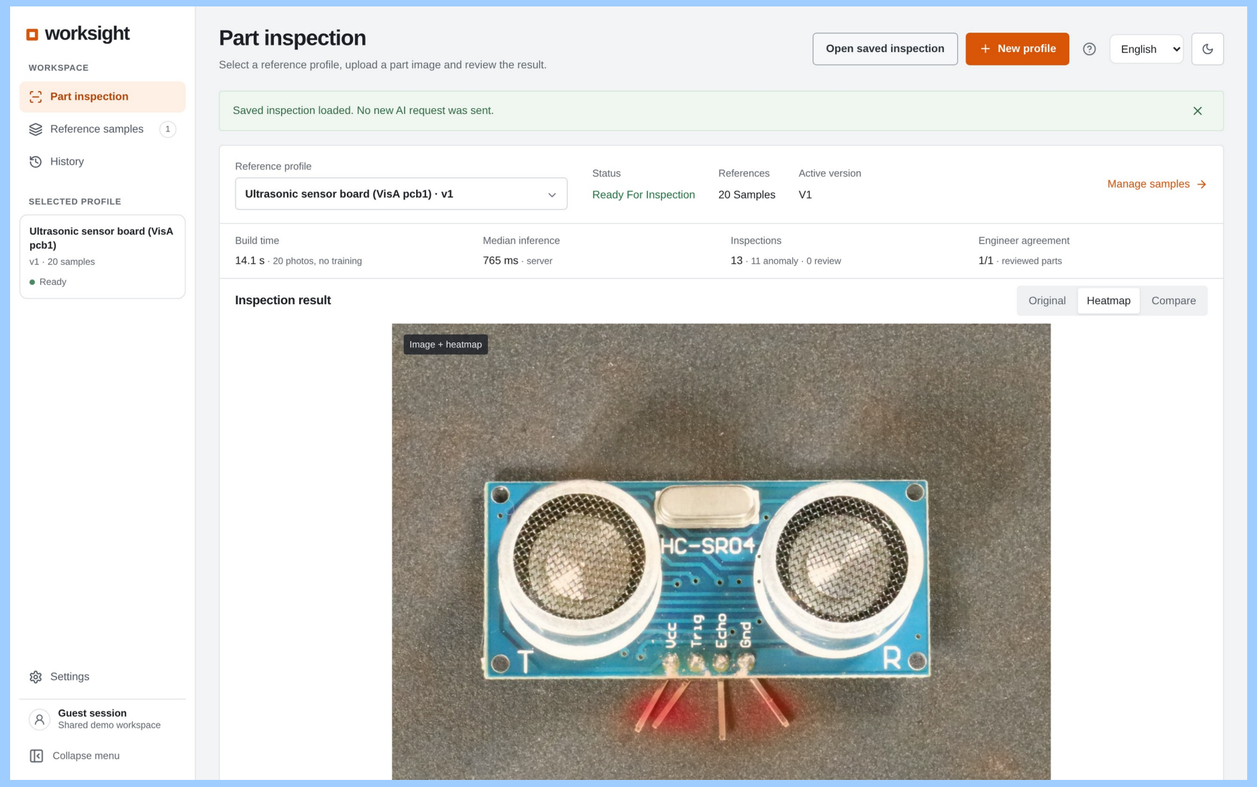
How it works

1. 20 photos of good parts build a profile in about 9 seconds. No defect labels, no training.

2. Each new part gets an anomaly score, a heatmap of where it differs and the closest good reference.

3. Gemma on AMD Token Factory explains what looks wrong and what to check. The engineer records the decision.





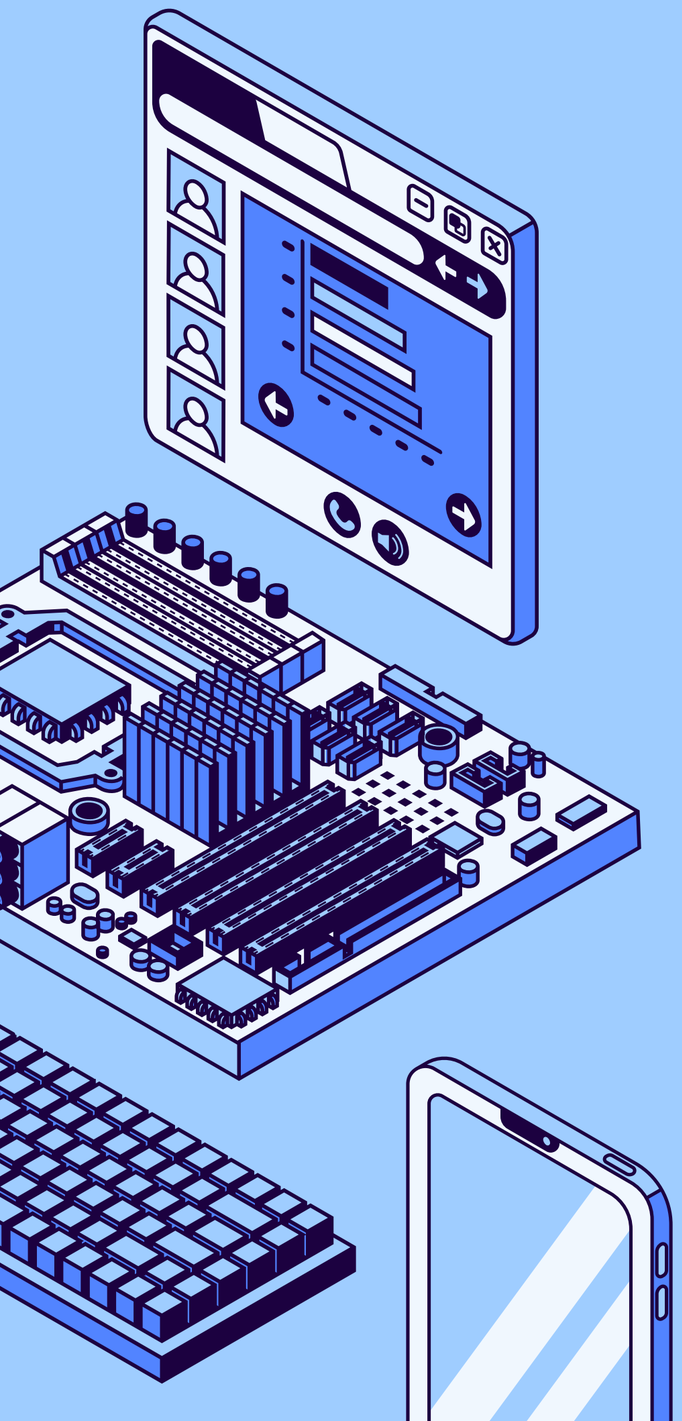
Algorithm, AI and engineer in one view.

Heatmap and closest good reference · Pass, Fail or Needs review

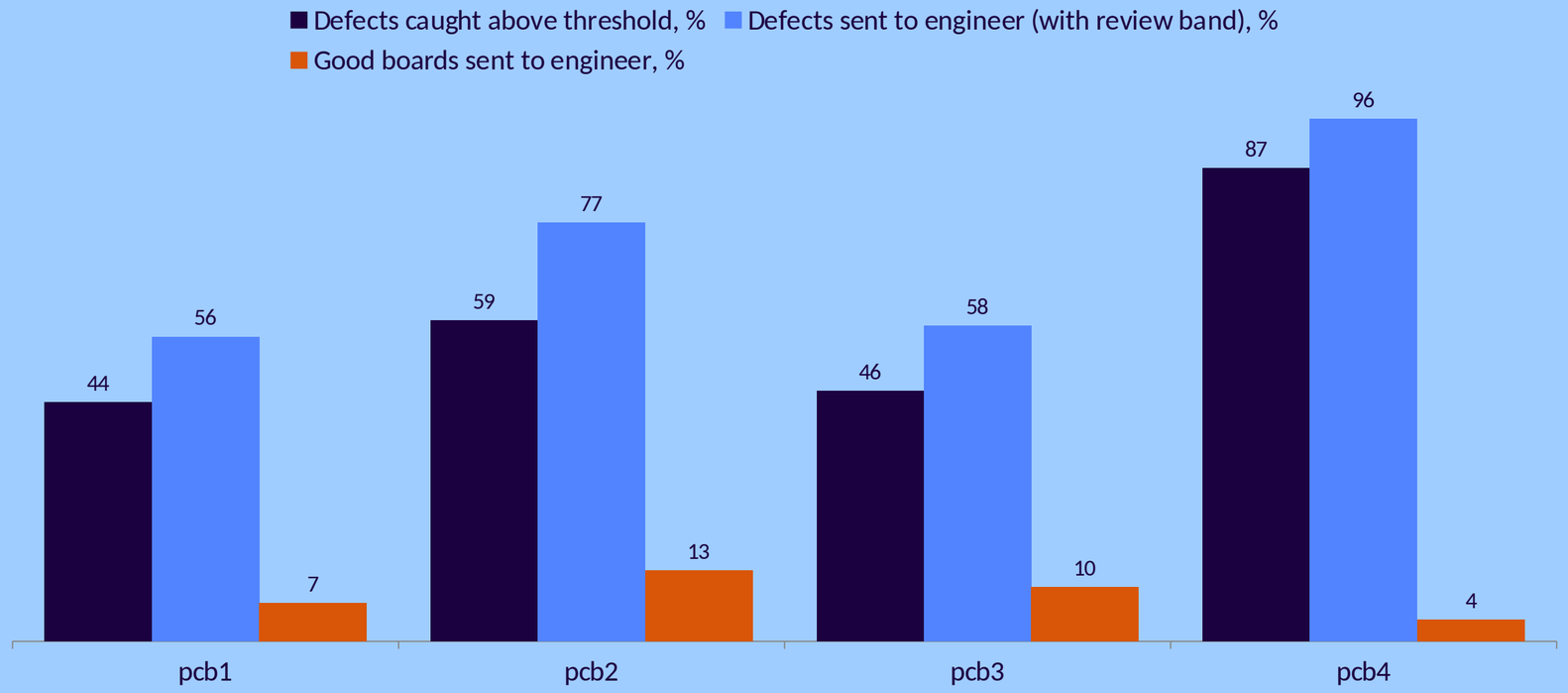
Gemma's explanation starts with each inspection · one-page PDF report

History, batch inspection, part documents · 7 languages · guided tour

The live product



Benchmarks



Public VisA dataset, four circuit-board sets, held-out test: 100 defective and about 100 good boards each, 20 good references per profile, CPU. Not an AMD GPU run.

AUROC 0.88–0.99 on four PCB sets.

Detector v2 raised dev AUROC from 0.806 to 0.862 (384 px, all patches).

A 24-variant scoring sweep found nothing better, so production stayed unchanged.

0.19–0.30 s median per image on CPU.

Set	AUROC	F1	False alarms	Recall	p50, ms
pcb1	0.905	0.591	5%	44%	250
pcb2	0.884	0.715	6%	59%	304
pcb3	0.875	0.613	4%	46%	250
pcb4	0.992	0.921	2%	87%	194

Same references and test as v1; numbers from docs/BENCHMARK.md in the repository.

Analytics



Small, fast and easy to inspect.

Detector: PyTorch, frozen ResNet-18 features, nearest-patch distance, threshold from leave-one-out scores

Backend: FastAPI, SQLite records, background job queue, PDF/DOCX export

Web: Next.js 16, React, TypeScript, zod

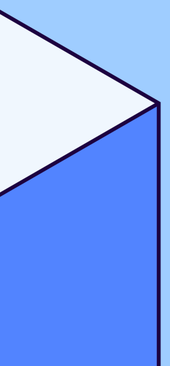
AI: Gemma-4-31B-IT on AMD Token Factory, DeepSeek-V4.1-Flash fallback

Hosting: Railway · Tests: pytest, Vitest, Playwright

Tech stack







Partner tech and why

AMD Token Factory

Runs Gemma on AMD GPUs behind an OpenAI-compatible API. The key stays on our server.

Gemma 4 31B

Google's vision-language model reads the photo, heatmap, closest reference and part documents and explains in plain words.

DeepSeek V4.1

Fallback model on Token Factory, so an explanation still arrives when Gemma is busy.

AMD Instinct

Next: the detector itself on an AMD GPU with ROCm. The run script is ready in the repo.



Market and competition

AI visual inspection is forecast to grow by about $27B by 2029 (Technavio).

Most tools need labelled defects, or are code libraries for data scientists.

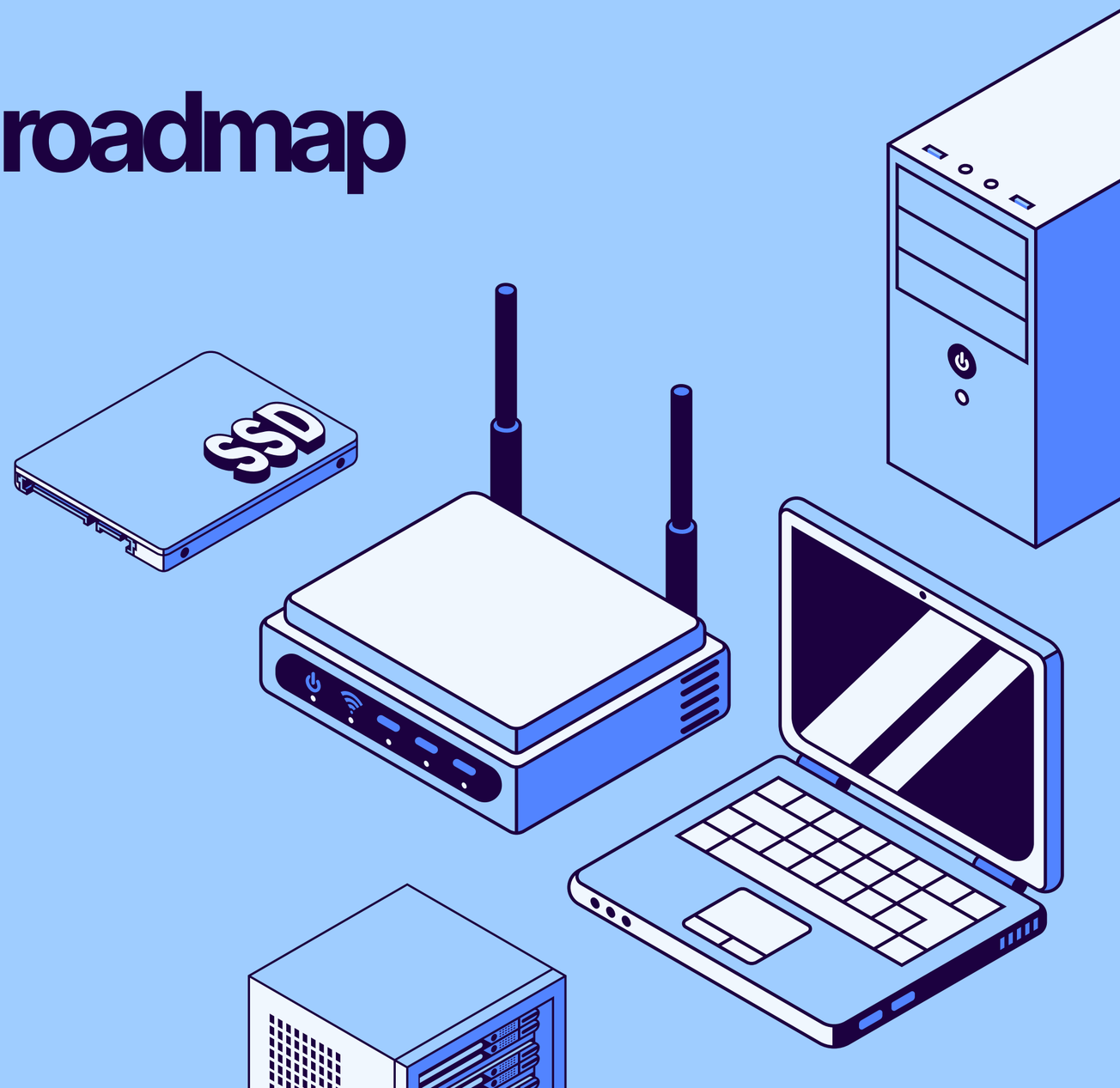
Worksight starts from good photos, explains each result in plain words and keeps the engineer's decision separate: a workspace for the line, not a notebook.

Business model and roadmap

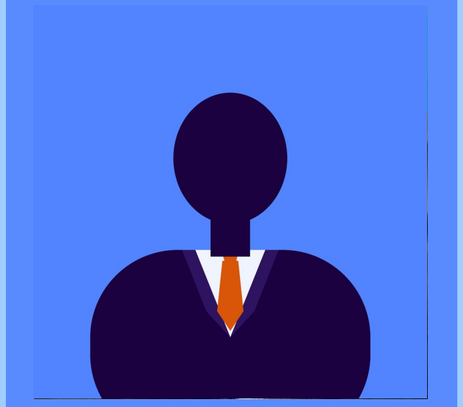
1. Subscription per camera view, or on-premises on AMD hardware for plants that keep images on site.

2. Pilot on one line and one camera: measure engineer minutes per part and escaped defects.

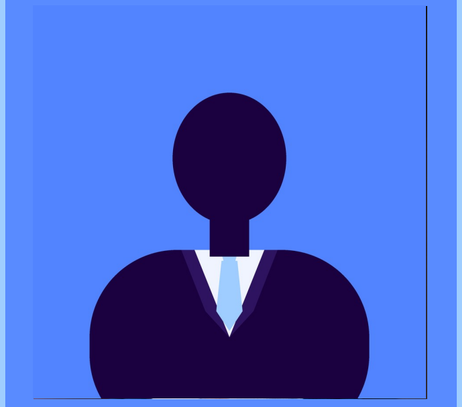
3. Next: detector on AMD Instinct, personal accounts and team workspaces, edge station on AMD Ryzen AI.



About us · Desert Eagles



Bakhtiyar
CEO · AI engineer



Ali
Frontend developer

Two-person team behind Worksight.

Bakhtiyar: detector, benchmarks, Gemma integration.

Ali: operator workspace and reports.

Telegram: @kingcrimson314





Try it

Live demo

web-production-d9743.up.railway.app

GitHub

github.com/DesertEaglesTech/worksight

Telegram

t.me/kingcrimson314

Worksight

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Thank You

